Nonlinear PCA and Manifold Learning

Reading: Izenman 2008: Chapter 16.1-16.2; 16.5-16.6

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Agenda

Motivation

Nonlinear PCA

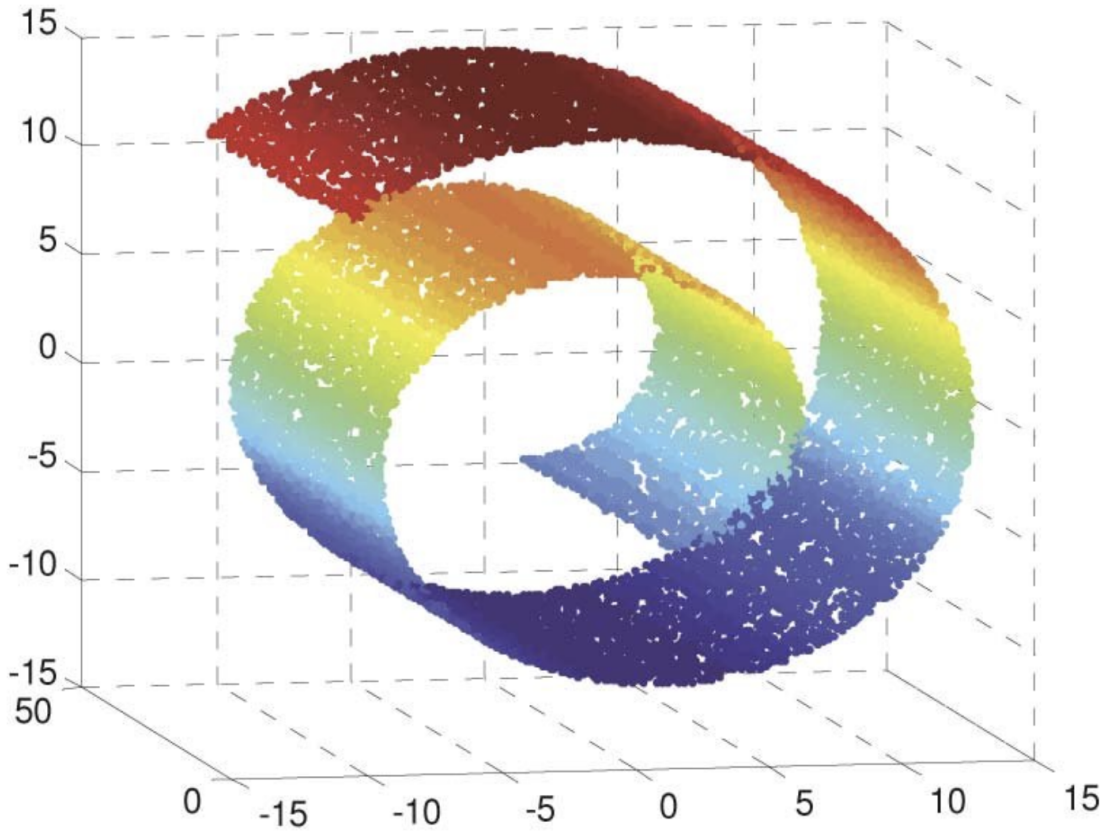
Manifold Learning

Motivation: Beyond Linear Dimensionality Reduction

- High-dimensional data may lie on a lower-dimensional, nonlinear manifold.
- **Ambient dimension:** Dimension of the space containing the data.
- **Intrinsic dimension:** Number of coordinates needed to describe the manifold.
- **Goal:** Find a low-dimensional representation that preserves the manifold's underlying geometry.

Swiss Roll: Why Geometry Matters

- A two-dimensional surface embedded in three-dimensional space.
- Points close in Euclidean distance may be far apart *along the surface*.
- **Key idea:** Recover the intrinsic geometry rather than the surrounding Euclidean geometry.



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- Understand the motivation behind manifold learning
- Describe key nonlinear dimensionality reduction techniques
- Implement and interpret nonlinear PCA methods, Isomap, LLE, and t-SNE in `R`
- Compare manifold methods to Linear PCA

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- There is no single unique "nonlinear PCA" – different methods generalize different properties of PCA

Polynomial PCA: Basic Idea

- **Idea:** Transform the original variables into polynomial features, then apply ordinary PCA.

- For $X = (X_1, \ldots, X_p)^\top$, a quadratic feature expansion is

$$\phi(X) = (X_1, \ldots, X_p, X_1^2, \ldots, X_p^2, X_1X_2, \ldots, X_{p-1}X_p)^\top.$$

- Apply PCA to the centered feature vectors $\phi(X)$:
  - Leading eigenvectors identify directions of maximum variation in feature space.
  - Small eigenvalues may reveal approximate polynomial constraints.

Nonlinear relationships become linear relationships in the expanded feature space.

Polynomial PCA: Interpretation and Limitations

- **Example:** Suppose $Y = X^2 + \varepsilon$, where ε represents small noise.
  - Expand to $\phi(X, Y) = (X, Y, X^2)^\top$.
  - The nonlinear relationship becomes $Y - X^2 \approx 0$, a linear constraint in feature space.

- **Two interpretations:**
  - Large eigenvalues: dominant variation in the expanded space.
  - Small eigenvalues: approximate polynomial constraints.

- **Advantages:** Simple; captures polynomial nonlinear relationships.

- **Limitations:** Rapid feature growth, numerical instability, and possible overfitting.

Feature expansion does not guarantee a useful low-dimensional manifold embedding.

Kernel PCA: PCA in Feature Space

- **Idea:** Apply linear PCA after mapping observations into a nonlinear feature space:

$$\Phi : \mathbb{R}^p \to \mathcal{H}.$$

- Instead of explicitly computing $\Phi(x)$, use a kernel function:

$$K(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle_{\mathcal{H}}.$$

- The kernel allows PCA to be performed using only pairwise similarities between observations.

Kernel PCA = linear PCA in an implicitly defined feature space.

Kernel PCA: Algorithm

1. **Choose a kernel**, e.g., the Gaussian (RBF) kernel:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right).$$

2. **Construct and center** the $n \times n$ Gram matrix:

$$K_{ij} = K(x_i, x_j), \qquad \widetilde{K} = HKH,$$

   where $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$.

3. **Compute principal-component scores** from the eigendecomposition:

$$\widetilde{K} v_\ell = \mu_\ell v_\ell, \qquad z_\ell = \sqrt{\mu_\ell}\, v_\ell.$$

The leading eigenvectors provide a nonlinear representation of the observations.

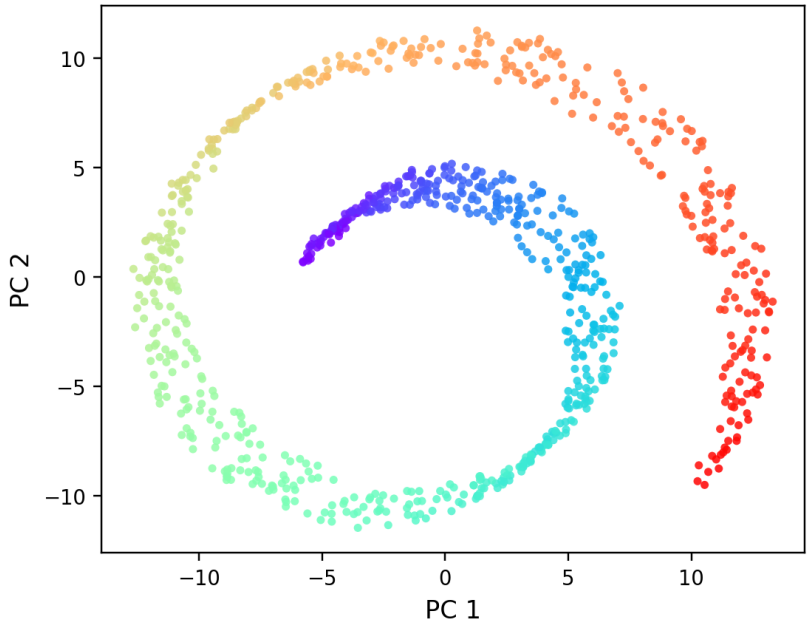
Kernel PCA: Interpretation and Limitations

- **What does kernel PCA do?**
  - Finds directions of maximum variance in the kernel-defined feature space.
  - Can reveal nonlinear structure missed by ordinary PCA.
- **Kernel choice matters:**
  - Different kernels define different notions of similarity.
  - For the RBF kernel, bandwidth σ controls the scale of similarity.
- **Limitation:** Kernel PCA does not explicitly preserve geodesic distances or guarantee recovery of intrinsic manifold coordinates.

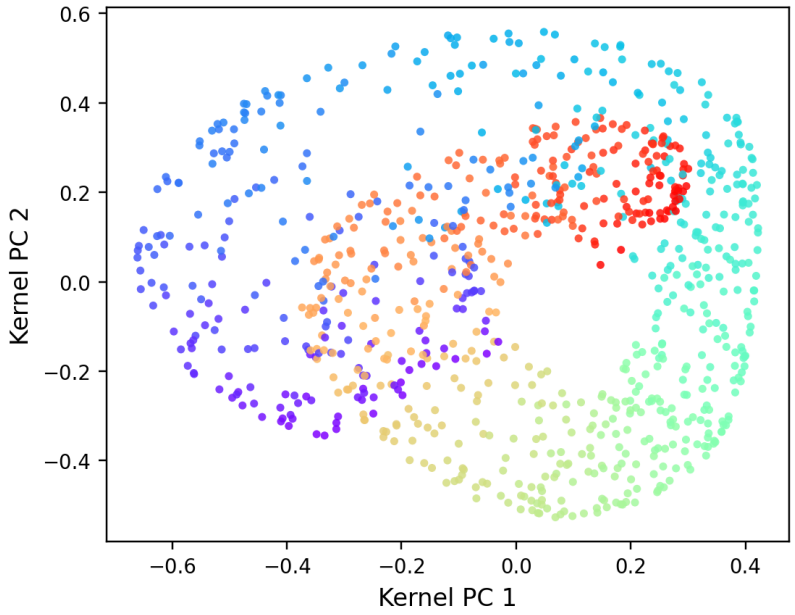
Nonlinear feature extraction $\neq$ manifold unfolding.

Linear PCA vs. Kernel PCA: Swiss Roll

Linear PCA



Kernel PCA (RBF)



- Projects onto linear subspace.
- May overlap distinct parts of the curved manifold.
- Extracts nonlinear features.
- Embedding depends on the kernel bandwidth.

Kernel PCA captures nonlinear variation, but does not necessarily unfold the manifold

Takeaways: Nonlinear PCA

- **Linear PCA:** Finds directions of maximum variance in the original space; may miss nonlinear structure.

- **Polynomial PCA:**
  - Applies PCA to explicitly constructed polynomial features.
  - Simple conceptually, but feature dimension grows rapidly.

- **Kernel PCA:**
  - Performs PCA in an implicit feature space using a kernel.
  - Avoids explicit feature expansion, but requires an $n \times n$ kernel matrix.

Key idea: Nonlinear features enable nonlinear representations, but do not guarantee manifold unfolding.

Manifold Learning: Core Idea

- High-dimensional data may lie on a low-dimensional nonlinear manifold.
- **Goal:** Find a low-dimensional embedding that reveals the underlying geometric structure.
- **Key idea:** Exploit relationships among nearby observations rather than relying only on global Euclidean distances.
- **Methods:**
  - Isomap and LLE: Geometry-based embeddings.
  - t-SNE and UMAP: Neighborhood-similarity embeddings.

Isomap: Preserving Geodesic Distances

- **Idea:** Replace Euclidean distances in classical MDS with approximate geodesic distances along the manifold.

- **Algorithm:**

  1. Construct a weighted neighborhood graph (k-NN or ε-neighborhood).

  2. Compute shortest-path distances between all pairs.

  3. Apply classical MDS to the resulting distance matrix.

- **Preserves:** Approximate global geodesic distances.

- **Limitation:** Sensitive to graph shortcuts and disconnected neighborhoods.

Locally Linear Embedding (LLE)

- **Idea:** Approximate each observation by a weighted combination of its nearest neighbors.

- **Algorithm:**
  1. Identify the k-nearest neighbors of each observation.
  2. Estimate weights that reconstruct each observation from its neighbors.
  3. Find a low-dimensional embedding that preserves these reconstruction weights.

- **Preserves:** Local reconstruction relationships.

- **Limitation:** Sensitive to neighborhood selection, noise, and unstable reconstruction weights.

t-SNE (t-distributed Stochastic Neighbor Embedding)

- Primarily used for **nonlinear visualization** in two or three dimensions.
- Converts pairwise distances into **similarity probabilities** in both high- and low-dimensional spaces.
- Finds an embedding by minimizing the **Kullback–Leibler (KL) divergence** between these distributions.
- **Interpretation:** Nearby points often represent similar observations, but cluster sizes, distances between clusters, and axis directions should not be interpreted literally.

UMAP (Uniform Manifold Approximation and Projection)

- Primarily used for **nonlinear visualization** and dimensionality reduction.
- Constructs a **weighted neighborhood graph** to represent local relationships among observations.
- Finds a low-dimensional embedding that approximately preserves this neighborhood structure.
- Often computationally efficient for large data sets; the embedding depends on neighborhood and distance parameters.

Interpretation: Local neighborhoods are generally more meaningful than global distances or axis directions.

Different Goals of Nonlinear Dimensionality Reduction

- **Polynomial PCA / Kernel PCA:**
  - Capture variance in transformed feature spaces.
  - Do not explicitly preserve manifold geometry.
- **Isomap / LLE:**
  - Preserve geodesic distances (Isomap) or local reconstruction relationships (LLE).
  - Aim to recover aspects of manifold geometry.
- **t-SNE / UMAP:**
  - Emphasize local similarities or neighborhoods.
  - Primarily useful for nonlinear visualization.

Choosing a Dimensionality Reduction Method

Method	Main Objective	Best Suited For
PCA	Maximize retained variance	Linear structure and interpretable components
Kernel PCA	Maximize variance in feature space	Nonlinear feature extraction
Isomap	Preserve geodesic distances	Global manifold geometry
LLE	Preserve local reconstruction relationships	Local manifold structure
t-SNE	Preserve local similarities	Exploratory visualization
UMAP	Preserve weighted neighborhood structure	Visualization and nonlinear embeddings

Key takeaway: Choose the method based on what structure you want to preserve.